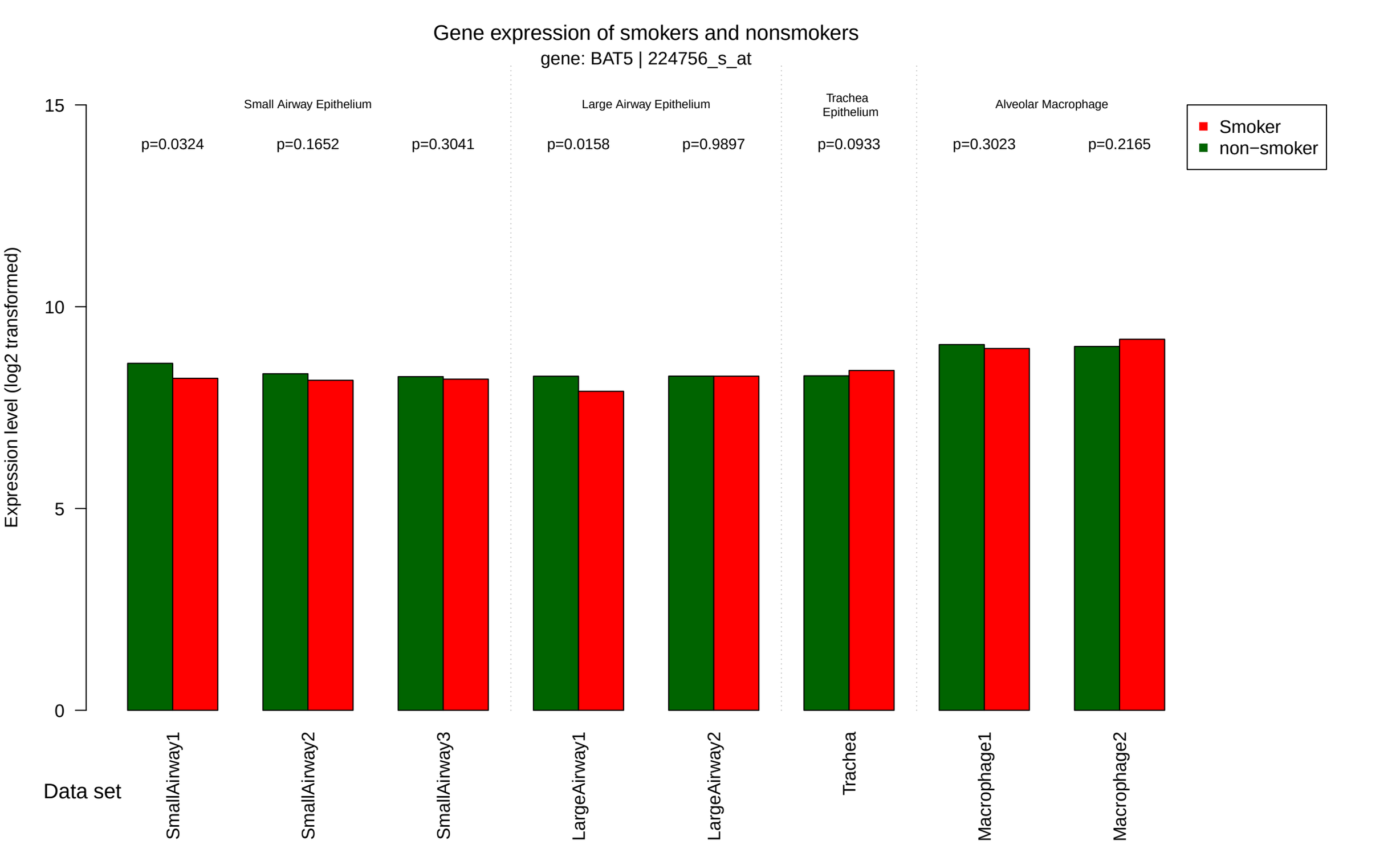
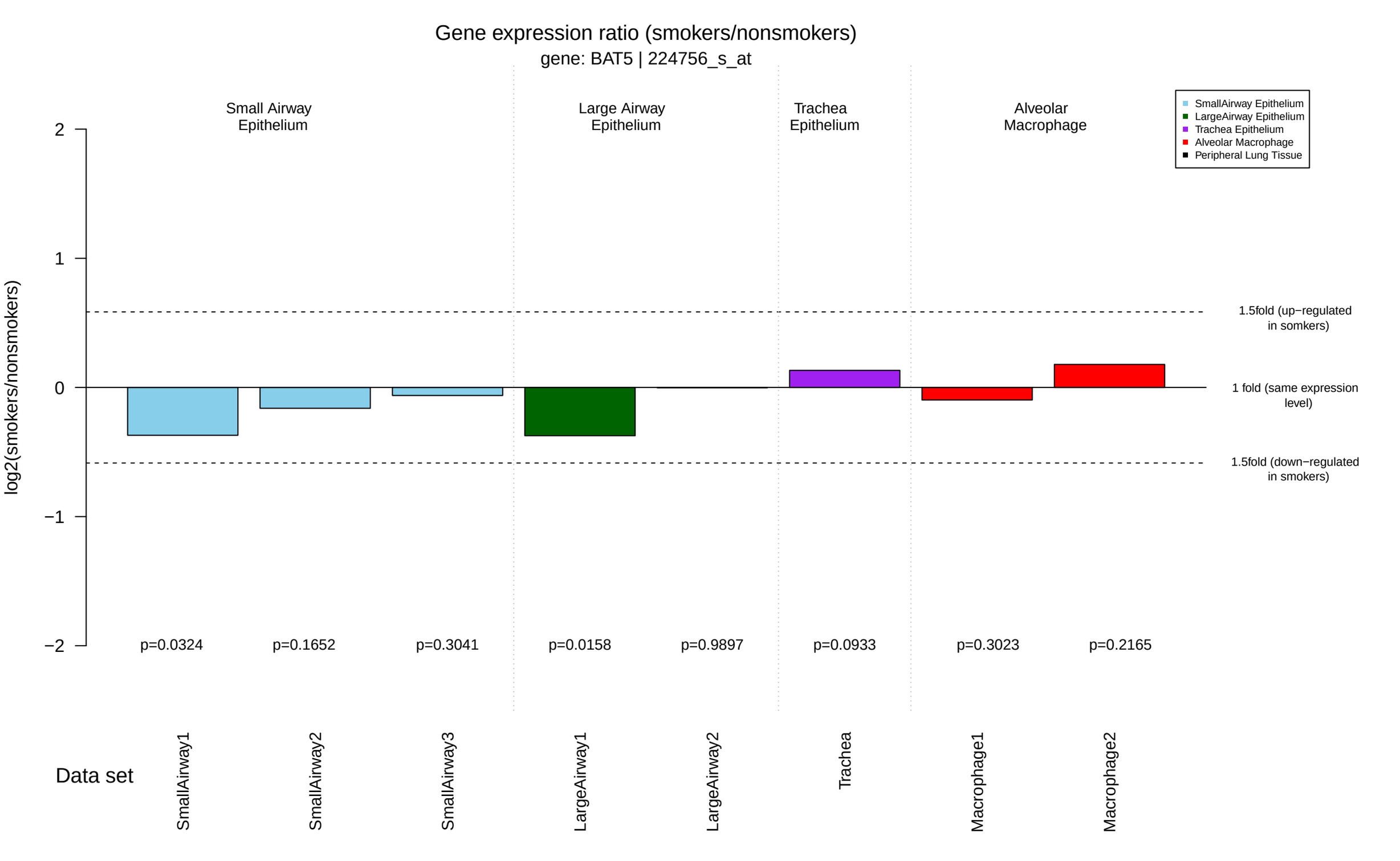
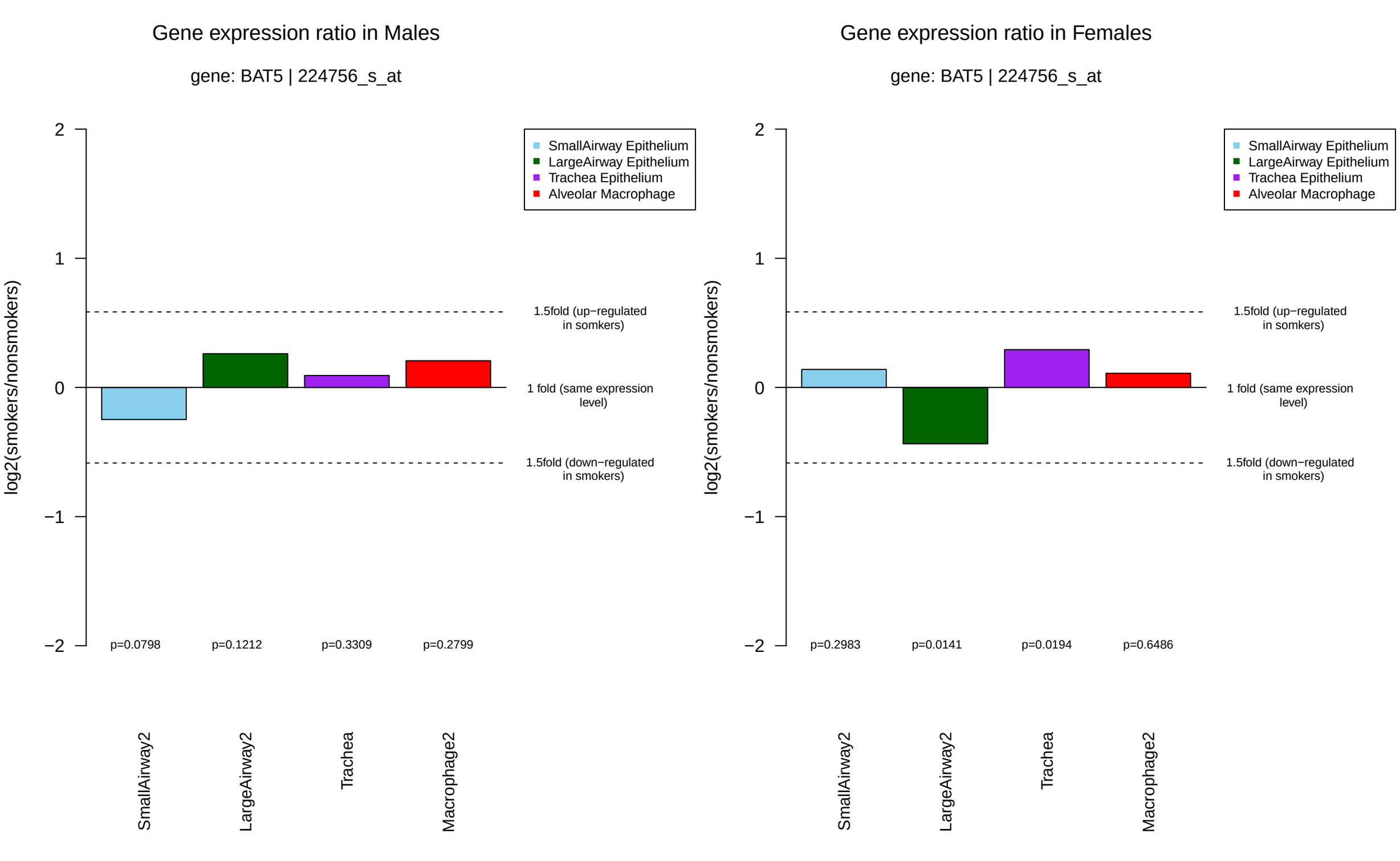
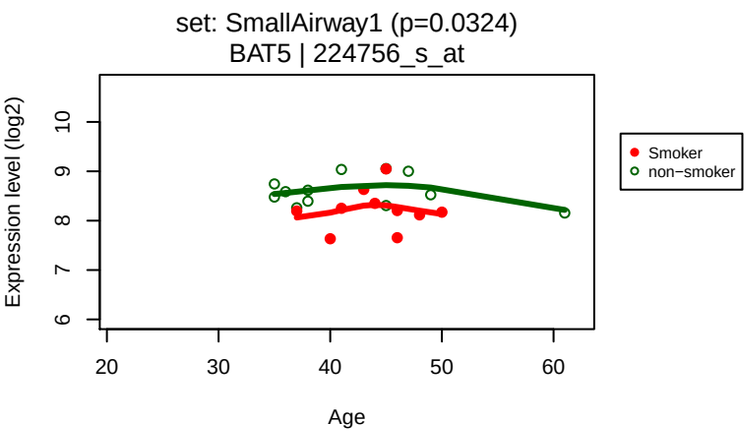
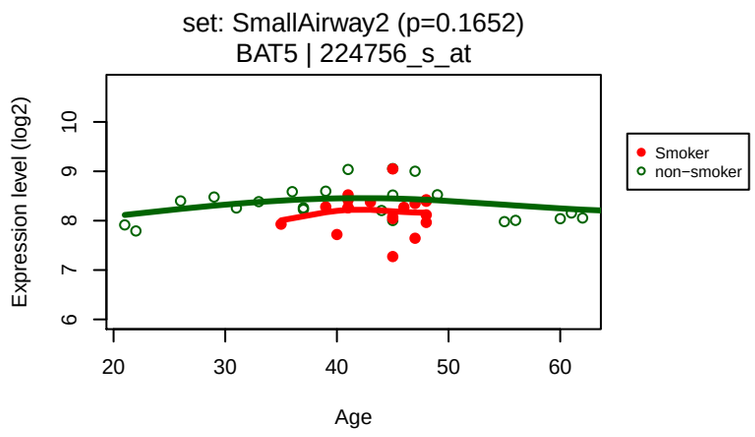
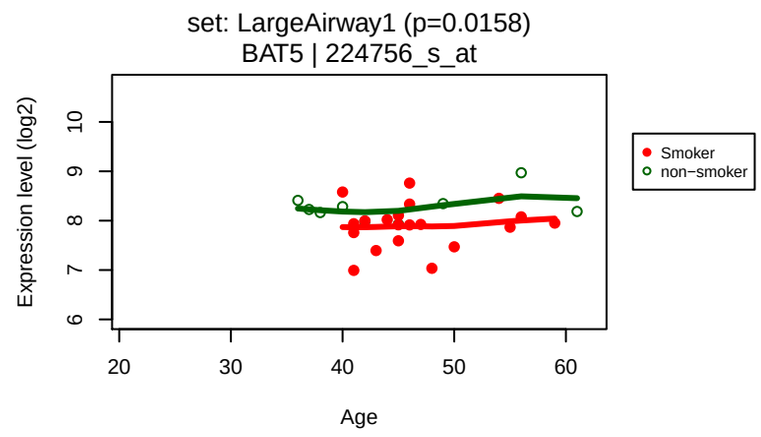

Gene expression of smokers and nonsmokers
gene: BAT5 | 224756_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0324
p=0.1652
p=0.3041
p=0.0158
p=0.9897
p=0.0933
p=0.3023
p=0.2165
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

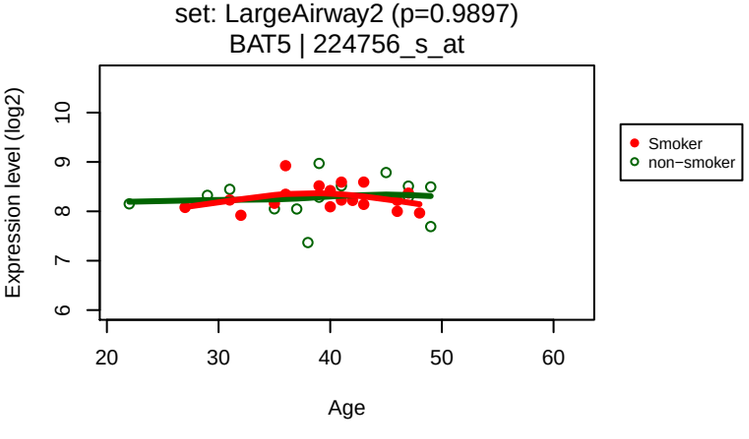

Gene expression ratio (smokers/nonsmokers)
gene: BAT5 | 224756_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0324
p=0.1652
p=0.3041
p=0.0158
p=0.9897
p=0.0933
p=0.3023
p=0.2165
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

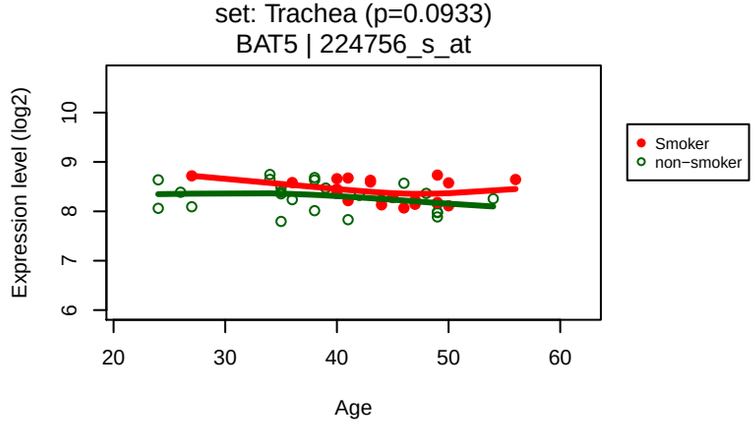

Gene expression ratio in Males
gene: BAT5 | 224756_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0798
p=0.1212
p=0.3309
p=0.2799
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BAT5 | 224756_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2983
p=0.0141
p=0.0194
p=0.6486
SmallAirway2
LargeAirway2
Trachea
Macrophage2

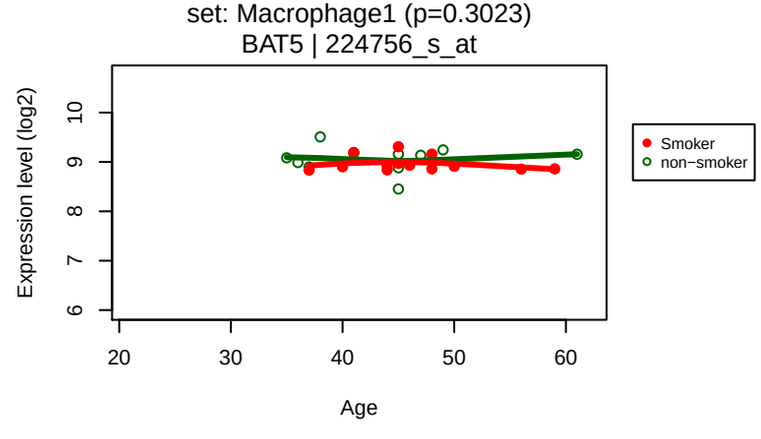

set: SmallAirway1 (p=0.0324)
BAT5 | 224756_s_at
Expression level (log2)
Age
Smoker
non−smoker

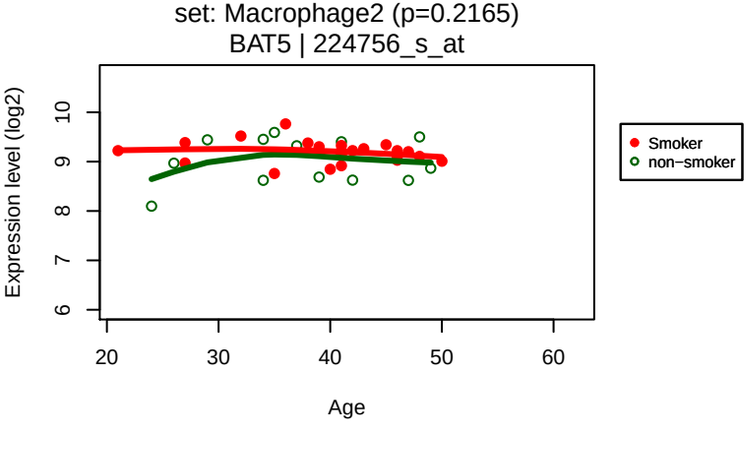

set: SmallAirway2 (p=0.1652)
BAT5 | 224756_s_at
Expression level (log2)
Age
Smoker
non−smoker

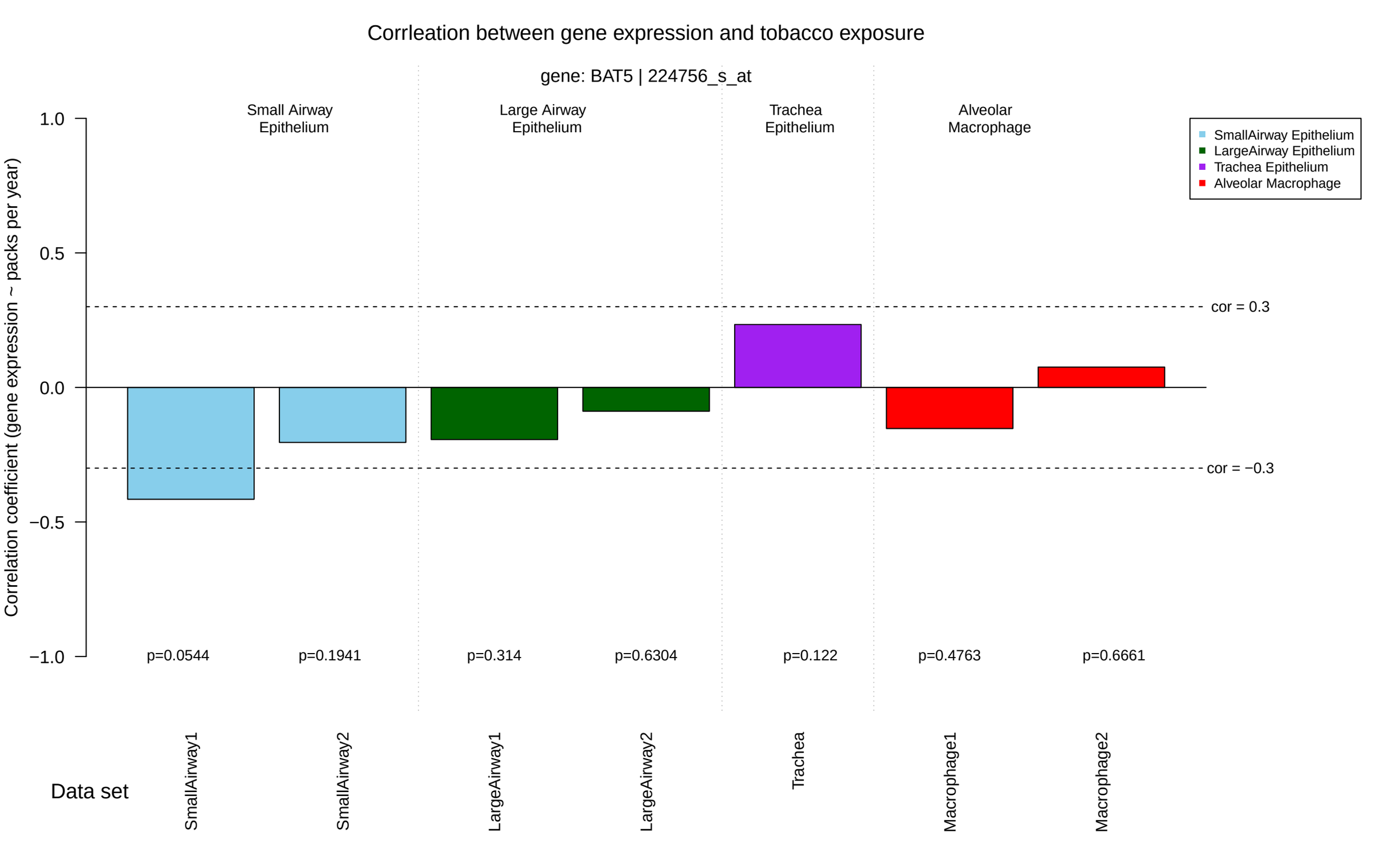

set: LargeAirway1 (p=0.0158)
BAT5 | 224756_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9897)
BAT5 | 224756_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0933)
BAT5 | 224756_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3023)
BAT5 | 224756_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2165)
BAT5 | 224756_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BAT5 | 224756_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0544
p=0.1941
p=0.314
p=0.6304
p=0.122
p=0.4763
p=0.6661
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2